

HOMEWORK 1
BIOSTATISTICS 794
DUE SEPTEMBER 14TH, 2025

Instructions:

- Show your work clearly and provide R code for all analyses.
- You may work with classmates to discuss approaches, but your write-up and code must be your own.
- Submit a Word file or knitted R Markdown (HTML or PDF).
- Include both text and figures/plots where appropriate.

1. For each situation below, state whether the analysis is **supervised** or **unsupervised** learning. Also classify the data structure as **wide**, **large**, **skinny**, or **messy**. Briefly explain your reasoning.
 - a) (5 points) You have survey data on 500 patients with their responses to 40 lifestyle questions, and you want to group them into lifestyle “clusters.”
 - b) (5 points) You have 2000 patients with blood pressure, BMI, and smoking status recorded, and you want to predict who will develop heart disease within 5 years.
 - c) (5 points) You have a genomic dataset with 100 patients and 15,000 gene expression measures, and you want to predict drug response.
 - d) (5 points) You have electronic health records (EHR) for 800 patients from a very busy ER in a large metropolitan area. The data are available for various markers and ICD-9 codes, but are spotty. You want to build a model to predict the length of stay.
2. We will use the `birthwt` dataset from the `MASS` package. This dataset contains information on the birth weights of infants and risk factors such as mother’s age, weight, smoking status, and medical history.

Below is some code to get you started with this data:

```
# Load and prepare data
data("birthwt", package = "MASS")
# Remove "low" variable.
bw <- birthwt[,-1]

# Create factors where appropriate
bw$race <- factor(bw$race, labels = c("white", "black", "other"))
bw$smoke <- factor(bw$smoke, labels = c("nonsmoker", "smoker"))
bw$ht <- factor(bw$ht, labels = c("no", "yes"))
```

```

bw$sui      <- factor(bw$sui, labels = c("no", "yes"))

# Define outcome and predictors
outcome <- "bwt"
predictors <- setdiff(names(bw), outcome)

# Generate all subsets up to size 5 (to keep it manageable)
subsets <- unlist(
  lapply(1:4, function(k) combn(predictors, k, simplify = FALSE)),
  recursive = FALSE
)

# Create formulas
formulas <- lapply(subsets, function(vars) {
  as.formula(paste(outcome, "~", paste(vars, collapse = " + ")))
})

```

This code will generate the formulas for all the models you need to fit for a, d, and e below. For b and c, you will only give the list of predictor variables. For these questions, include all predictor variables.

- a) (10 points) Fit a standard multiple linear regression with **birth weight** as the outcome. Use adjusted R^2 , BIC, and Mallows Cp to explore which predictors are important.
- b) (5 points) Use backwards selection criteria to find the best model.
- c) (5 points) Fit a **forward stagewise regression** model on the same data.
- d) (10 points) Perform 10-fold **cross-validation** to evaluate predictive error. Instead of mean squared error, use the **Median Absolute Deviation (MAD)**:

$$MAD = \text{median}(|y_i - \hat{y}_i|).$$

MAD is sometimes preferred to MSE when there are outliers in the data. MAD is less sensitive to outliers than MSE is.

- e) (15 points) Perform either:
 - **Nested CV**, or
 - **CV with a separate test set**
 to both select the best model *and* to estimate its prediction error (still using MAD).
- f) (5 points) Compare results across all approaches and write a short paragraph on which model you would choose and why.

3. We will use the `airquality` dataset (built into R), which records daily air quality measurements in New York, May–September 1973.

- a) (5 points) Fit a linear regression model predicting **Ozone** levels using predictors you think are appropriate (e.g., Solar, Wind, Temp, Month). You don't need to perform any model selection; simply select a model that you think is appropriate and use it for the remainder of this question.
- b) (15 points) Check the key regression assumptions:
- Linearity (residual plots, component+residual plots),
 - Homoscedasticity (residuals vs fitted, Breusch–Pagan test),
 - Normality of residuals (QQ plot),
 - Influence (Cook's distance, leverage).
- c) (10 points) Estimate the **prediction error (MAD)** of your model separately for each **Month**. Report MAD in May, June, July, August, and September. Comment on whether the model performs equally well across months, and what that might suggest about seasonal variation.